A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, III
Revision Log: Draft 9 to Draft 10 (exposition only)

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Scope

The third exposition pass requested by David Joyner, in the layout of Part II (his choice): Summary of results after §1, notation and glossary and related work at the back. **No mathematical change**: no statement, proof, hypothesis or label was changed. Part III's own numbering in §§1–12 is unchanged; the old §13 ("Selected terminology and notation"), which no other paper cites, becomes §13 "Notation and glossary", and a new §14 "Related work" follows it. This is the last separate draft of Part III: by David's decision the four Parts and the companion paper are now being combined into one book, and the separate drafts are frozen.

Changes

1. **References to Part I** renumbered to Part I, Draft 11 (the reordered draft): Lemma 2.2 → 4.2, Definition 4.1 → 6.1, Proposition 2.3 → 4.3, Proposition 3.5 → 5.5 (six places), Theorem 3.9 → 5.9; thirteen occurrences in all, twelve after "Part I" and one unprefixed ("treated there in Proposition 3.5"). References to Part II (Definition 6.4, Corollary 6.8) are unchanged, since Part II's numbering did not change. The new §1 lead-in names the drafts cited (Part I, Draft 11; Part II, Draft 10).

2. **(a) Glossary.** §13: lead-in, `\input{../glossary}`, and §13.11 "Notation and terminology specific to Part III", which keeps every entry of the old §13 except "Good element" (now in the shared glossary) and adds entries for H, Z, $\widetilde{H}$, $\widetilde{Z}$, $\widetilde{T}$, H^1, U_D^1, tame/wild, the Hilbert spaces of §4, b_D, c_D and the character matrix, $\mathcal{C}_{\eta^{-1}}(H)$ and $L_D^{(\widetilde{\eta})}$, and Ψ_D and the target identity (38).

3. **(b) Summary of results.** New unnumbered section after §1: eleven items with labels and pointers, plus the running example.

4. **(c) Related work.** New §14, referring to Part I, §2, for the series and adding what is specific to Part III: the division-algebra side (DKV, Badulescu, Cohen; Riehm on $\mathrm{SL}_1(D)$; Prasad–Raghunathan on its topological central extensions; Klose's local paper), power operations (Adams, Atiyah, Segal), metaplectic correspondences (Flicker–Kazhdan, Zou), and arithmetic (Kottwitz, Milne, the Grunwald–Wang theorem, Loeser). Four references added: Artin–Tate (checked: W. A. Benjamin, 1968; Chapter X is "The Grunwald–Wang theorem", per the AMS table of contents of the 2009 edition); Klose, Math. Z. 191 (checked on EuDML); Prasad–Raghunathan,

Invent. Math. 92 (1988), and Riehm, Amer. J. Math. 92 (1970) (bibliographic data checked only through Prasad–Rapinchuk's reference list on numdam). The originals of the last three were not consulted; the text cites them for their subject only.

5. **(d) Lead-ins.** §1 has a new lead-in paragraph; the lead-ins of §§2–5 and 7–12 now say what each section depends on and what notation it introduces; §6 and §10 are marked as background or exposition. The labels paragraph points to the shared definitions.

6. Title line: Draft 10, November 8, 2026.

Point for David

Proposition 7.2 (unique splitting over H^1 when $p \nmid n$) and Proposition 8.1 (non-splitting when $p \mid n$) concern the same objects as Prasad–Raghunathan's work on topological central extensions of $\mathrm{SL}_1(D)$ and Klose's local paper. A comparison would be a useful check; it has not been made.

Files

`part-III/mjl_part3_draft10.tex` and `.pdf`; this log (`.tex`, `.pdf`); `README.md`. Draft 9 is kept unchanged.